Video content (5 minutes)

1) 30second clip of just the camera + gps
2) 30second clip of just the mpu6050 + bmp280
3) 30 second screen recording of the website
4) 30 second long clip for (1)(2)(3)(4)
5) Discuss about vega as flight controller and flight computer both and future plans (technical side) 2 minute
6) Discuss some business potential (mention costings and market) 1 minute

Total : 5 minutes

```
mpuvideosim                                   100%  466KB  33.4KB/s   00:13
amrut@Maverick:~$ scp -O root@192.168.1.20:/root/mpu_flight.csv /home/amrut
root@192.168.1.20's password:
mpu_flight.csv                                100%   33KB  31.4KB/s   00:01
amrut@Maverick:~$ vim bmp-video-simulation.c
amrut@Maverick:~$ /home/amrut/VEGA/trisul32_buildroot/buildroot/output/host/bin/
riscv32-linux-gcc /home/amrut/bmp-video-simulation.c -o bmpvideosim -static -lm
amrut@Maverick:~$ /home/amrut/VEGA/trisul32_buildroot/buildroot/output/host/bin/
riscv32-linux-strip bmpvideosim
amrut@Maverick:~$ scp -O bmpvideosim root@192.168.1.20:/root/
root@192.168.1.20's password:
bmpvideosim                                   100%  458KB  33.3KB/s   00:13
amrut@Maverick:~$
```

7)

Flow to Interface the Camera Sensor

- Interface the camera sensor correctly (see below) keeping the board off

To interface ArduCam

cs	c1	J6 pin 5
mosi	h1	J6 pin 4
miso	g1	J6 pin 1
sck	f1	J6 pin 3
gnd	Gnd	J7 gnd
vcc	3.3/5v	J7 vcc 3.3/5v
sda	sda	sda
scl	scl	scl

- Turn on the board
- Go to Vivado (/opt/Vivado/2023.2/bin/vivado) and flash with this bitstream : icarus v2.2.bit via hardware manager, ensure there is so negative slack
- Go to /home/amrut/VEGA/vega-tools/utils/eth_transfer and do ./send.sh bbl.bin riscv.dtb (basically the buildroot)
- Open fresh terminal and do minicom trisul32
- Bootloader… do root x 2 and continue
- Set up eth0 and enp1s0 everytime

On board :

```
ip link set eth0 up

ip addr add 192.168.1.20/24 dev eth0
```

On Laptop :

```
sudo ip addr flush dev enp1s0

sudo ip addr add 192.168.1.10/24 dev enp1s0

sudo ip link set enp1s0 up
```

- Write the required code in c, and to access the address, review the image in the next page…. (since camera is using i2c + spi, the image only describes flow for giving address for spi)
- Scp from laptop the compiled capture binary (from /home/amrut/Downloads)

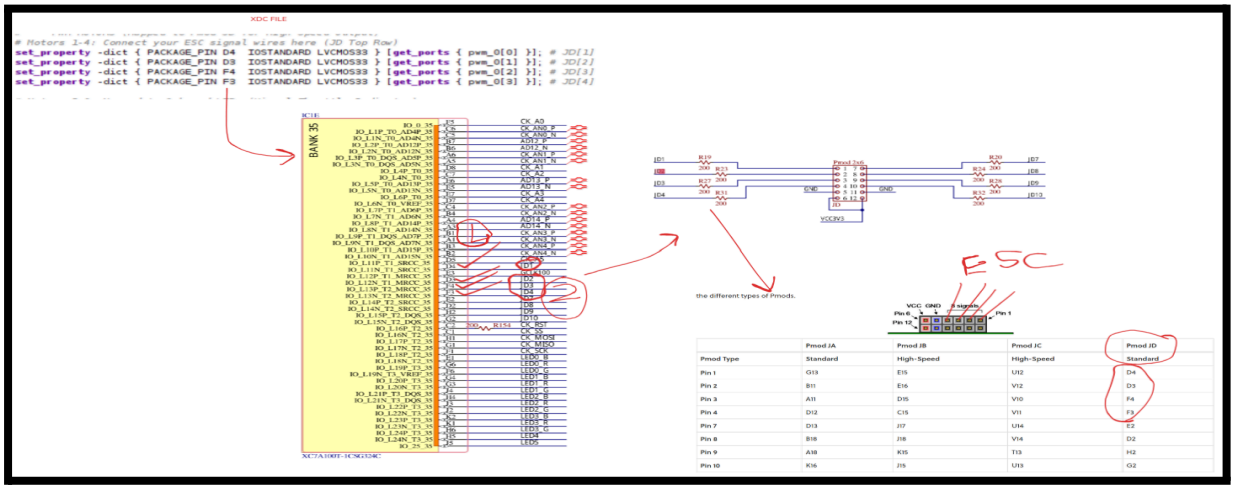
```
/home/amrut/VEGA/trisul32_buildroot/buildroot/output/host/bin/riscv32-linux-gcc /home/amrut/Downloads/<whatever>.c -o <whatever> -static

/home/amrut/VEGA/trisul32_buildroot/buildroot/output/host/bin/riscv32-linux-strip <whatever>

ssh-keygen -f '/home/amrut/.ssh/known_hosts' -R '192.168.1.20'

scp -O <compiled_binary> root@192.168.1.20:/root/
```

- Run the binary (ensure you strip the debug headers)
- SCP the image.bin back
- Run the python script as provided in /camera in git

Check if /dev/ttyS*

Check tty for current tty

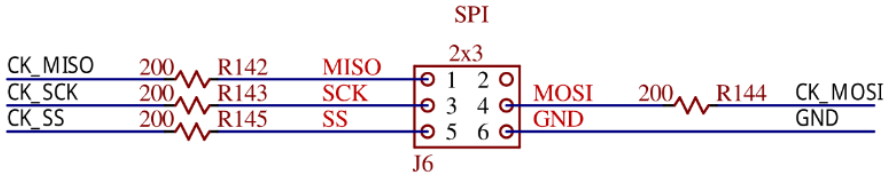
To interface the camera sensor



To interface ArduCam

cs	c1	J6 pin 5
mosi	h1	J6 pin 4
miso	g1	J6 pin 1
sck	f1	J6 pin 3
gnd	Gnd	J7 gnd
vcc	3.3/5v	J7 vcc 3.3 -> 5v
sda	sda	sda
scl	scl	scl

```
# --- SPI 1 (Camera/Aux) -> Mapped to Standard ChipKit SPI Header ---

set_property -dict { PACKAGE_PIN C1  IOSTANDARD LVCMOS33 } [get_ports { spi_rtl_1_ss_io[0] }]; # CK_SS

set_property -dict { PACKAGE_PIN H1  IOSTANDARD LVCMOS33 } [get_ports { spi_rtl_1_io0_io }];   # CK_MOSI

set_property -dict { PACKAGE_PIN G1  IOSTANDARD LVCMOS33 } [get_ports { spi_rtl_1_io1_io }];   # CK_MISO

set_property -dict { PACKAGE_PIN F1  IOSTANDARD LVCMOS33 } [get_ports { spi_rtl_1_sck_io }];   # CK_SCK
```



Why the board might not have internet……

Reasons:

1. FPGA Linux has **no Ethernet interface (`eth0`)**
2. No DHCP client
3. No default route
4. No DNS
5. Probably **no SSH server running**

Extra Information :) ---->

(In hardware and software, a magic number is a special, fixed value used for identification or control, often appearing as unexplained literals in code (e.g., 37px for positioning) or as file signatures (e.g., %PDF for PDFs) that uniquely identify file types or protocols, helping systems quickly recognize data formats. While useful for defining system parameters, they can make code harder to maintain, so they are often replaced by named constants for clarity, especially in hardware design and digital forensics

PNG File: Starts with 89 50 4E 47 (hex) which is ASCII for ".PNG".

JPEG File: Starts with FF D8 FF E0 (hex).

Why They Matter

Identification: Quickly identify file types or data formats without parsing the whole file.

Control: Dictate system behavior, like clock speeds in hardware designs.

Security: Vital in cybersecurity for malware detection and file validation.

Maintainability: Poorly documented magic numbers make code confusing; defining them as constants improves clarity)

Flow to Interface the mpu6050

- Interface the camera sensor correctly (see below) keeping the board off

To interface mpu6050

gnd	Gnd	J7 gnd
vcc	3.3/5v	J7 vcc 3.3v (5v)
sda	sda	sda
scl	scl	scl

- Turn on the board
- Go to Vivado (/opt/Vivado/2023.2/bin/vivado) and flash with this bitstream : icarus v2.2.bit via hardware manager, ensure there is so negative slack
- Go to /home/amrut/VEGA/vega-tools/utils/eth_transfer and do ./send.sh bbl.bin riscv.dtb (basically the buildroot)
- Open fresh terminal and do minicom trisul32
- Bootloader… do root x 2 and continue
- Set up eth0 and enp1s0 everytime
    - On board :
    - ip link set eth0 up
    - ip addr add 192.168.1.20/24 dev eth0
    - On Laptop :
        - sudo ip addr flush dev enp1s0
        - sudo ip addr add 192.168.1.10/24 dev enp1s0
        - sudo ip link set enp1s0 up
- Write the required code in c (we are only using the i2c), and to access the address, (in mpu6050-addrmapping) review the image in the next page…. (since mpu is using i2c)
- Scp from laptop the compiled capture binary (from /home/amrut/Downloads)
    - /home/amrut/VEGA/trisul32_buildroot/buildroot/output/host/bin/riscv32-linux-gcc /home/amrut/Downloads/<whatever>.c -o <whatever> -static
    - /home/amrut/VEGA/trisul32_buildroot/buildroot/output/host/bin/riscv32-linux-strip <whatever>
    - ssh-keygen -f '/home/amrut/.ssh/known_hosts' -R '192.168.1.20'
    - scp -O <compiled_binary> root@192.168.1.20:/root/

Flow to Interface the Neo-6m-GPS

- Interface the gps sensor correctly (see below) keeping the board off

To interface gps

gnd	Gnd	J7 gnd
vcc	3.3/5v	J7 vcc 3.3/5v
Tx (on gps)	Uart-0-rx	JA PMOD JA1 (Pin 1) (Upper row, rightmost pin)
Rx (on gps)	Uart-0-tx	JA PMOD JA2 (Pin 2) (Upper row, one pin left of rightmost pin)

- Turn on the board
- Go to Vivado (/opt/Vivado/2023.2/bin/vivado) and flash with this bitstream : icarus v2.2.bit via hardware manager, ensure there is so negative slack
- Go to /home/amrut/VEGA/vega-tools/utils/eth_transfer and do ./send.sh bbl.bin riscv.dtb (basically the buildroot)
- Open fresh terminal and do minicom trisul32
- Bootloader… do root x 2 and continue
- Set up eth0 and enp1s0 everytime

On board :

```
ip link set eth0 up

ip addr add 192.168.1.20/24 dev eth0
```

On Laptop :

```
sudo ip addr flush dev enp1s0

sudo ip addr add 192.168.1.10/24 dev enp1s0

sudo ip link set enp1s0 up
```

- Write the required code in c, and to access the address

- Scp from laptop the compiled capture binary (from /home/amrut/Downloads)

  ```
  /home/amrut/VEGA/trisul32_buildroot/buildroot/output/host/bin/riscv32-linux-gcc /home/amrut/Downloads/<whatever>.c -o <whatever> -static
  ```

  ```
  /home/amrut/VEGA/trisul32_buildroot/buildroot/output/host/bin/riscv32-linux-strip <whatever>
  ```

  ```
  ssh-keygen -f '/home/amrut/.ssh/known_hosts' -R '192.168.1.20'
  ```

  ```
  scp -O <compiled_binary> root@192.168.1.20:/root/
  ```

- Run the binary (ensure you strip the debug headers)

Flow to Interface the bmp280

- Interface the camera sensor correctly (see below) keeping the board off

To interface mpu6050

gnd	Gnd	J7 gnd
vcc	3.3/5v	J7 vcc 3.3v (5v)
sda	sda	sda
scl	scl	scl

- Turn on the board
- Go to Vivado (/opt/Vivado/2023.2/bin/vivado) and flash with this bitstream : icarus v2.2.bit via hardware manager, ensure there is so negative slack
- Go to /home/amrut/VEGA/vega-tools/utils/eth_transfer and do ./send.sh bbl.bin riscv.dtb (basically the buildroot)
- Open fresh terminal and do minicom trisul32
- Bootloader… do root x 2 and continue
- Set up eth0 and enp1s0 everytime
    - On board :
    - ip link set eth0 up
    - ip addr add 192.168.1.20/24 dev eth0
    - On Laptop :
        - sudo ip addr flush dev enp1s0
        - sudo ip addr add 192.168.1.10/24 dev enp1s0
        - sudo ip link set enp1s0 up
- Write the required code in c (we are only using the i2c), and to access the address, (in mpu6050-addrmapping) review the image in the next page…. (since mpu is using i2c)
- Scp from laptop the compiled capture binary (from /home/amrut/Downloads)
    - /home/amrut/VEGA/trisul32_buildroot/buildroot/output/host/bin/riscv32-linux-gcc /home/amrut/Downloads/<whatever>.c -o <whatever> -static
    - /home/amrut/VEGA/trisul32_buildroot/buildroot/output/host/bin/riscv32-linux-strip <whatever>
    - ssh-keygen -f '/home/amrut/.ssh/known_hosts' -R '192.168.1.20'
    - scp -O <compiled_binary> root@192.168.1.20:/root/

Our dts (device tree) file

```
/dts-v1/;
/ {
  #address-cells = <2>;
  #size-cells = <2>;
  compatible = "ucbbar,spike-bare-dev";
  model = "ucbbar,spike-bare";
  chosen {
    stdout-path = &SERIAL0;
    bootargs = "console=hvc0 earlycon";
  };
  cpus {
    #address-cells = <1>;
    #size-cells = <0>;
     timebase-frequency = <10000>;
    CPU0: cpu@0 {
      device_type = "cpu";
      reg = <0>;
      status = "okay";
      compatible = "riscv";
      riscv,isa = "rv32ima";
      mmu-type = "riscv,sv32";
```

```
    clock-frequency = <40000000>;



CPU0_intc: interrupt-controller {
      #interrupt-cells = <1>;
      interrupt-controller;
      compatible = "riscv,cpu-intc";
    };
  };
};
memory@80000000 {
  device_type = "memory";
  reg = <0x0 0x80000000 0x0 0x10000000>;


};
soc {
  #address-cells = <2>;
  #size-cells = <2>;
  compatible = "ucbbar,spike-bare-soc", "simple-bus";
  ranges;
  clint@20010000 {
    compatible = "riscv,clint0";
    interrupts-extended = <&CPU0_intc 3 &CPU0_intc 7>;
```

```
  reg = <0x0 0x20010000 0x0 0xc0000>;

};

ethernet@20030000 {
        device_type = "network";
        compatible = "cdac,cdac_mac";
        interrupt-parent = <&plic0>;
        interrupts = <9>;
        reg = <0x0 0x20030000 0x0 0x100>;
    };

plic0: interrupt-controller@20010000 {
#interrupt-cells = <1>;
compatible = "cdac,plic-1.0.0";
reg = <0x0 0x20010000 0x0 0x4000000>;
riscv,ndev = <32>;
interrupt-controller;
interrupts-extended = <&CPU0_intc 9>;
};

SERIAL0: ns16550@10001000{
```

```
  compatible = "ns16550a";
  clock-frequency = <40000000>;
  reg = <0x0 0x10001000 0x0 0x100>;
  reg-io-width = <4>;
  reg-shift = <2>;
  current-speed = <115200>;

};
spi_clock:spi_clock{
        #clock-cells = <0>;
        compatible = "fixed-clock";
        clock-frequency = <40000000>;
    };
    spi0: spi@10000600 {
        compatible = "cdac,spi";
        clock-names = "spi_clock";
        clocks = <&spi_clock>;
        interrupt-parent = <&plic0>;
        interrupts = <6>;
        reg = <0x0 0x10000600 0x0 0x100 >;
        #address-cells = <1>;
        #size-cells = <0>;
        spidev@0 {
```

```
        compatible = "cdac,spidev";
        reg = <0>;
        spi-max-frequency = <25000000>;
    };
};
gpio0: gpio-controller@10080000 {
    compatible = "cdac,cdac-gpio";
    reg = <0x0 0x10080000 0x0 0x50000>;
    gpio-base = <0>;
    gpio-controller;
    #gpio-cells = <2>;
};
gpio1: gpio-controller@10180000 {
    compatible = "cdac,cdac-gpio";
    reg = <0x0 0x10180000 0x0 0x50000>;
    gpio-base = <16>;
    gpio-controller;
    #gpio-cells = <2>;
};
i2c0: i2c@10000800 {
            compatible = "cdac,mdp-i2c";
        clock-frequency = <100000>;
        system-frequency = <40000000>;
```

```
        #address-cells = <1>;
        #size-cells = <0>;
        reg = <0x0 0x10000800 0x0 0x100>;
        interrupt-parent = <&plic0>;
        interrupts = <8>;
        status = "okay";
};
    i2c1: i2c@10000900 {
        compatible = "cdac,mdp-i2c";
        clock-frequency = <100000>;
        system-frequency = <40000000>;
        #address-cells = <1>;
        #size-cells = <0>;
        reg = <0x0 0x10000900 0x0 0x100>;
        interrupt-parent = <&plic0>;
            interrupts = <4>;
        status = "okay";

};
pwm_clock:pwm_clock{
    #clock-cells = <0>;
    compatible = "fixed-clock";
    clock-frequency = <40000000>;
```

```
        };
        pwm0: pwm@10400000 {
            compatible = "cdac,cdac-pwm";
            clock-names = "pwm_clock";
                clocks = <&pwm_clock>;
            reg = <0x0 0x10400000 0x0 0x100>;
            #pwm-cells = <3>;


        };
    };
    };
```